INCH-POUND
MIL-DTL-24643/86
w/AMENDMENT 1
14 April 2020
SUPERSEDING
MIL-DTL-24643/86
29 October 2019

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 5,000 VOLTS, TYPE LS5KVTSSGUSHR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

The following commercial specifications shall be used in conjunction with this military specification to the extent indicated herein: UL 1072, ANSI/NEMA WC 53/ICEA T-27-581, ICEA T-24-380, ICEA T-32-645, and IEEE 1580.

Construction (Watertight with Circuit Integrity)

First	-	Copper-stranded conductor, coated or uncoated, with extruded or pumped waterblocking material (see table I for size).
Second	-	Semi-conducting fabric separator tape, applied helically with a minimum of 25 percent overlap.
Third	-	Extruded silicone rubber insulation, minimum average thickness of 0.050 inch.
Fourth	-	Silicone rubber/fiberglass tape insulation, minimum average thickness of 0.080 inch.
Fifth	-	Black semi-conducting layer consisting of two or more cross-lapped black semi-conducting fabric tapes. Each tape shall be helically applied with 40 percent minimum overlap. Total combined minimum average thickness of 0.010 inch. An alternate may consist of a black semi-conducting extruded compound layer, minimum average thickness of 0.010 inch.

NOTE 1: The combined extruded plus taped insulation wall thicknesses shall be 0.140 inch, minimum average. (The minimum/maximum insulation walls of UL 1072 are not applicable.)

NOTE 2: Identification of phase conductors shall be as follows: the underlined letters “A”, “B”, and “C” shall be printed on the component black semi-conducting layer. The letters shall be approximately 0.1875 inch high with alternate legends inverted and shall be printed at intervals not exceeding 1.5 inches. All three-phase conductors shall be printed. The letter axis may be perpendicular or parallel to the longitudinal axis of the component and shall be printed in white ink.

The opposite side of each component, 180 degrees from the identification letter code, shall be printed with “SEMI-CONDUCTING COATING - REMOVE BEFORE TERMINATING OR SPLICING.”

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The letters shall be approximately 0.1875 inch high with alternate legends inverted and shall be printed at intervals not exceeding 12 inches. All three-phase conductors shall be printed. The letter axis shall be perpendicular to the longitudinal axis of the component and shall be printed in white ink.

Sixth - The three conductors shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm, well-rounded assembly.

Seventh - Semi-conducting fabric tape applied helically with 25 percent minimum overlap.

Eighth - Braided shield of tin-coated copper, 34 AWG, 95 percent minimum coverage. Shield shall have a 0.0163 square inch (10.5 square millimeters) minimum copper area.

Ninth - Polyester tape applied helically with 25 percent minimum overlap.

Tenth - Extruded low smoke, zero halogen, cross-linked polyolefin jacket. Minimum average wall thickness of 0.090 inch.

NOTE: A ripcord and manufacturer's identification tape (see 3.4.9 of MIL-DTL-24643), both laid longitudinally under the cable jacket, are required.

TABLE I. Cable details.

Military part number M24643/86	Type and size	Conductor sizes		Cable overall diameter max. (inches)	Conductor resistance (ohms)	Insulation resistance (megohms)	Accelerated service loading (amperes)
		Navy STD	AWG/ MCM				
-01UO	LS5KVTSSG USHR-100		100 (class D)	1.870	0.106	400	295
-02UO	LS5KVTSSG USHR-250		250 (class C)	2.300	0.0453	300	535
-03UO	LS5KVTSSG USHR-400	400 (127)		2.750	0.0273	250	750

REQUIREMENTS:

Qualification required.

INSPECTION:

Partial discharge (PD) level (corona) test: The finished cable shall be measured for partial discharge extinction level in accordance with ICEA T-24-380. The inception voltage shall be equal to the voltage at the minimum acceptable partial discharge extinction level specified below. A Hipotronics partial discharge detector model DDX 7000, or equal, shall be utilized.

Acceptance criteria: During each test, there shall be no evidence of arcing or corona discharge within the insulation system of each conductor. The extinction level, when defined as the voltage at which the partial discharge is less than 5 picocoulombs, shall be 5 kilovolts (kV). In the event that the inception is not reached and the 5 kV is attained, the test shall be terminated. In such a case, the cable shall be considered partial discharge resistant and as having passed the test.

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Minimum bending radius: The term "minimum bend radius" indicates the radius of the smallest diameter circle around which the cable may be bent, resulting in the maximum amount of cable bend allowed. Permanent training bend radius applies to cable being bent in place, with no tension applied to the cable or conductors. Pulling/installing bend radius applies to cable being pulled over sheaves, or around raceway/conduit bends, with tension applied to the cable or conductors.

Permanent training:

100 MCM size	14.5 inches, No damage
250 MCM size	17.5 inches, No damage
400 MCM size	20.5 inches, No damage

Pulling/Installing:

100 MCM size	21.5 inches, No damage
250 MCM size	26.5 inches, No damage
400 MCM size	32.0 inches, No damage

Dimensional measurements: This test shall be performed optically in accordance with ANSI/NEMA WC 53/ICEA T-27-581.

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	13,500
Conductor to black semi-conducting layer	Not applicable
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	See table I
Conductor to overall shield	225
Conductor and shield continuity	No failure
Partial discharge, conductor to shield (5 kV AC, max.) (picocoulombs) (see above for test procedure/information)	5
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure

Group B:

Cold bending cable	No failure
Thermoset proof test (percent, max.)	

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Overall cable jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C])	Zero
Physicals (unaged)	
Overall cable jacket	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) air oven	
Overall cable jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (overall cable jacket) (cycles, min.)	125
Permanence of printing (cycles, min.) (component black semi-conducting layer)	25
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Overall shield (conformance to material, construction, and coverage)	No failure
Surface transfer impedance	
Milliohms per meter (max.)	Not applicable
EMP response (dB, min.)	Not applicable

Group D:

Flame propagation (cable)	No failure
Gas flame (circuit integrity) (1 hour)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; Groups A, B, C, and D; plus the following:

Salt water immersion: This test shall determine the ability of the completed cable to withstand the effects of salt water and sea spray that may occur during marine service. The Salt Water Immersion Test detailed in IEEE 1580 shall be performed. Specimen test length shall be 8.0 feet (2.45 meters) in lieu of the 3.6 feet (1.1 meters) specified.

Acceptance criteria: After immersion in salt water, the cable shall conform to the following:

a. The cable shall pass the voltage withstand test of MIL-DTL-24643.

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b. There shall be no degradation of the mechanical properties of the jacket or insulation such that they would crack when wound around a mandrel having a diameter equal to sixteen times the sample maximum specified outer diameter.

c. The insulation and jacket shall not degrade to the point where either will crack or separate from the cable during the conditioning or during the testing required in items (a) and (b) above.

Compatibility of semi-conducting materials	This test shall be performed in accordance with ICEA T-32-645.
Cold working (minus 68±3.6 °F [20±2 °C])	No damage
Gas flame (circuit integrity) (3 hours)	No failure
Accelerated service (see table I for load current)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (overall cable jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket (overall cable)	2
Fillers	2
Insulation	18
Semi-conducting materials	18
Halogen content (percent, max.)	
Jacket (overall cable)	0.2
Fillers	0.2
Insulation	0.2
Semi-conducting materials	0.2
Immersion (overall cable jacket)	
Elongation (percent of unaged, min.)	50
Tensile strength (percent of unaged, min.)	50
Smoke index (max.)	
Jacket (overall cable)	25
Fillers	45
Insulation	45
Semi-conducting materials	45
Toxicity index (max.)	
Jacket (overall cable)	5
Fillers	5
Insulation	1.5

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Semi-conducting materials	Not applicable
Durometer (overall cable jacket) – type A (hardness, min.)	80
Weathering (overall cable jacket)	No failure
Electrical moisture absorption	Not applicable

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS5KVTSSGUSHR, all sizes	1,000

AMENDMENT NOTATIONS. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2020-006)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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